# Practice workbook · Parallel and perpendicular lines: proving it or finding an unknown

Using the rule that parallel lines have equal gradients and perpendicular gradients multiply to -1, either to prove a relationship or to find an unknown coordinate or gradient.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-COO-05

# 1. Question 5 · part a · 3 marks

2020 NOV · Paper 01

Worked-example retry · Attempt part a only.

**5** The points $P$, $Q$, $R$ and $S$ have coordinates (4, 7), (3, 0), (10, 1) and (11, 8) respectively.

(a) Show, by calculation, that the lines $PR$ and $QS$ are perpendicular.

**(3)**

(b) Find the exact lengths of (i) $PR$ (ii) $QS$

**(2)**

(c) Find the area of the quadrilateral $PQRS$

**(2)**

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## 1. Question 5 · part a · 3 marks · Working and answer

Working and answer

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## 2. Question 10 · part c.i · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part c.i only.

**10** The equation of the line $L_1$ is $y - 3x + a = 0$

The point $A$ with coordinates $(a, 10)$ lies on $L_1$

(a) Show that $a = 5$

**(2)**

Line $L_1$ crosses the $y$-axis at the point $B$

(b) Write down the coordinates of $B$

**(1)**

The point $C$ with coordinates $(50, -5)$ lies on $L_2$

Given that $L_2$ passes through $A$

(c) (i) show that $L_1$ and $L_2$ are perpendicular

**(3)**

(ii) hence find an equation for $L_2$ giving your answer in the form $y = px + q$

**(2)**

The point $D$ has coordinates $(m, n)$

The length of $CD$ is $15\sqrt{10}$ and the gradient of $BD$ is $-3$

(d) Find the value of $m$ and the value of $n$

**(6)**

(e) Find the area of quadrilateral $ABCD$

**(3)**

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## 2. Question 10 · part c.i · 3 marks · Working and answer

Working and answer

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## 3. Question 4 · part d.ii · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

**4** The point $A$ has coordinates $(-4, -10)$ and the point $B$ has coordinates $(3, 11)$
The line $l$ passes through $A$ and $B$.

(a) Find an equation of $l$.

**(2)**

The point $P$ lies on $l$ such that $AP : PB = 3 : 4$

(b) Find the coordinates of $P$.

**(2)**

The point $Q$ with coordinates $(m, n)$, where $m < 0$, lies on the line through $P$ that is perpendicular to $l$.

Given that the length of $PQ$ is $\sqrt{10}$

(c) find the coordinates of $Q$.

**(6)**

The point $R$ has coordinates $(-11, -21)$

(d) Show that

(i) $AB$ and $RQ$ are equal in length,

(ii) $AB$ and $RQ$ are parallel.

**(4)**

(e) Find the area of the quadrilateral $ABQR$.

**(2)**

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## 3. Question 4 · part d.ii · 4 marks · Working and answer

Working and answer

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## 4. Question 2 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

**2** The point $A$ has coordinates $(-7, -1)$ and the point $B$ has coordinates $(3, 4)$

(a) Find an equation of the line that passes through $A$ and $B$
Give your answer in the form $ax + by + c = 0$ where $a$, $b$ and $c$ are integers.

**(3)**

The point $C$ has coordinates $(-3, 7)$

Given that $k$ is a constant such that $AB = kAC$

(b) find the value of $k$

**(2)**

The point $D$ has coordinates $(3, p)$ where $p$ is a constant.

Given that $CD$ is perpendicular to $AB$

(c) find the value of $p$

**(3)**

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## 4. Question 2 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

**Question 2 continued**

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**(Total for Question 2 is 8 marks)**

## 4. Question 2 · part c · 3 marks · Working and answer

Working and answer

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## 5. Question 11 · part b · 7 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

**11** The points $A$ and $B$ have coordinates $(-3, -5)$ and $(7, 5)$ respectively.

(a) Find an equation for the line $AB$

**(2)**

The point $C$ has coordinates $(p, 1)$ where $p < 0$

Given that $AC$ and $BC$ are perpendicular,

(b) prove that $p = -5$

**(7)**

The point $D$, where $BCD$ is a straight line, is such that $C$ divides $BD$ in the ratio $4:3$

(c) Find the coordinates of $D$

**(2)**

(d) (i) Find the exact length of $AC$

(ii) Hence, or otherwise, find the area of triangle $ABD$

**(4)**

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## 5. Question 11 · part b · 7 marks · Working and answer

Working and answer

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# 6. Question 2 · part c · 8 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

**2** The point $A$ has coordinates $(3, 2)$, the point $B$ has coordinates $(8, 3)$ and the point $C$ has coordinates $(4, 7)$

(a) Show that $ABC$ is an isosceles triangle.

**(2)**

The midpoint of $BC$ is $M$

(b) Find an equation of the line that passes through $A$ and $M$

Give your answer in the form $y = mx + c$

**(3)**

The points $A$, $C$ and $D$ are collinear such that $AD = kAC \quad (k > 1)$

Given that $\angle ABD = 90^\circ$

(c) find the coordinates of $D$
Show your working clearly.

**(8)**

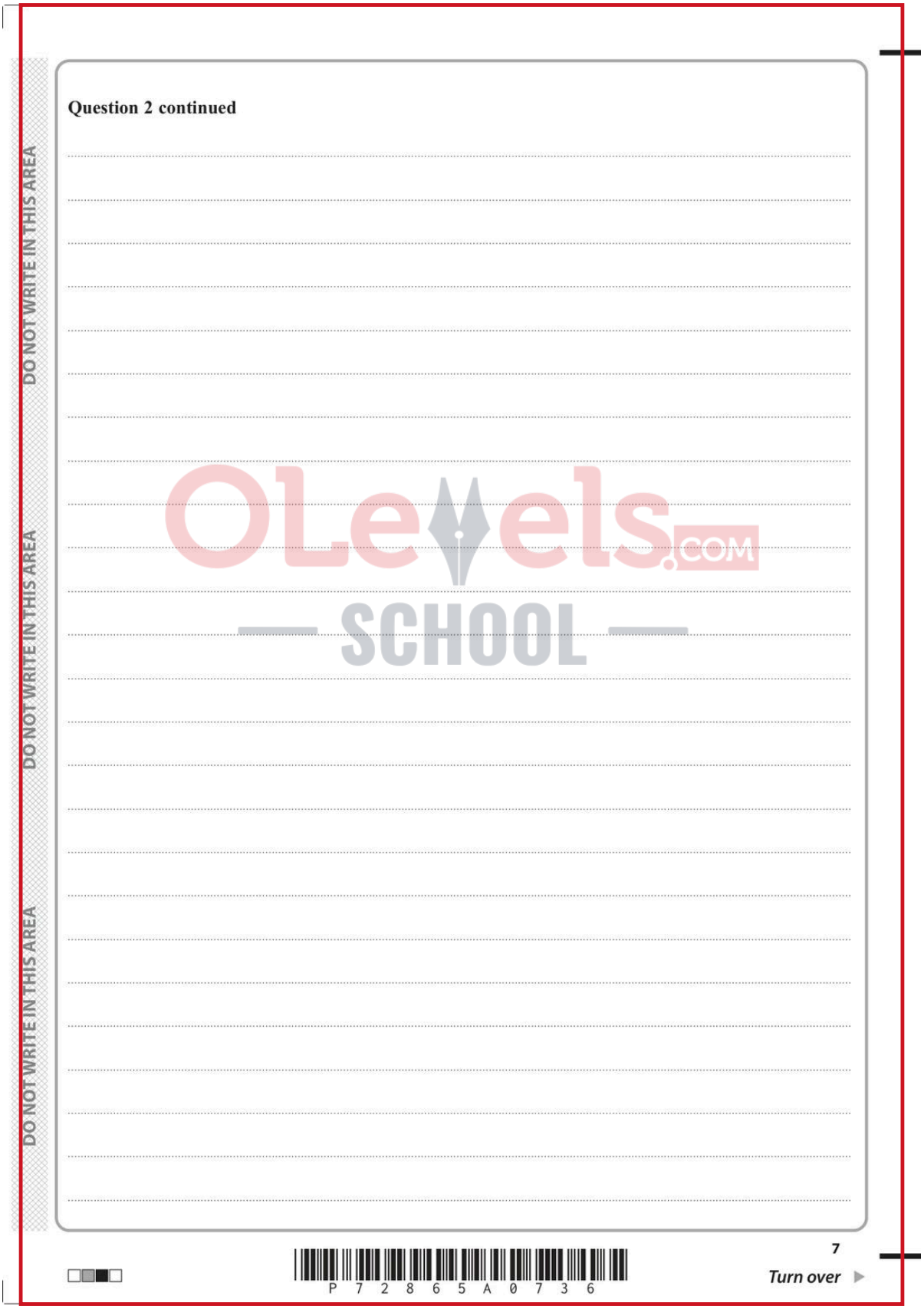
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## 6. Question 2 · part c · 8 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.



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**Question 2 continued**

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## 6. Question 2 · part c · 8 marks

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Further practice · Attempt part c only.

**Question 2 continued**

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## 6. Question 2 · part c · 8 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

**Question 2 continued**

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**(Total for Question 2 is 13 marks)**

## 6. Question 2 · part c · 8 marks · Working and answer

Working and answer

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## 7. Question 9 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

**9** The straight line $L_1$ passes through the point $A$ with coordinates (4, 7) and has gradient $m$, where $m < 0$

Another straight line $L_2$ is perpendicular to $L_1$ and passes through the point $B$ with coordinates $(4, k)$ where $k \neq 7$

The lines $L_1$ and $L_2$ intersect at the point $C$.

Given that the $y$ coordinate of $C$ is $Y$

(a) show that $Y = \dfrac{7 + m^2k}{m^2 + 1}$

**(7)**

Given that the triangle $ABC$ is isosceles,

(b) find the value of $m$

**(5)**

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## 7. Question 9 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

**Question 9 continued**

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## 7. Question 9 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

**Question 9 continued**

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## 7. Question 9 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.



**Question 9 continued**

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**(Total for Question 9 is 12 marks)**

## 7. Question 9 · part b · 5 marks · Working and answer

Working and answer

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